

Sensitivity and Design Tradeoffs for Flash Lidar Cameras

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In the design of any sensing device, the objective is to achieve a sufficient detection sensitivity for given operational constraints and finite allowances for size, weight, power, and cost. Key tradeoffs and fundamental limitations can be identified through the dependence of the sensitivity on design parameters. To define the sensitivity mathematically, it is necessary to apply domain knowledge about the nature of the signal that the device is intended to detect, the sources and magnitudes of undesirable inputs (i.e., noise), and the algorithm used to extract the signal.

This report presents a design tradeoff analysis for flash lidar cameras. The main result is an expression that relates the pixel signal-to-noise ratio (SNR) to the illumination output, pixel size, receiver lens characteristics, and the strength of background light. Derivation of the SNR is carried out under a set of simplifying assumptions that are clearly stated. The analysis is concluded with numerical estimates of various parameters for devices and conditions likely to be encountered in practice.

1 Flash Lidar Overview

At a high level, a flash lidar camera has three main components: 1) a narrowband, wide-beam illuminator (typically a laser) capable of producing very brief pulses, 2) a lens that focuses incoming light, and 3) a focal plane array (FPA) of light-sensitive detectors (pixels). Figure 1 shows an abstract view of the component arrangement as well as a projection of the FPA and the illuminated region (details are discussed in Section 3). These components are common to both flash lidar and conventional 2D cameras. What most distinguishes a flash lidar camera is that each pixel must record both time and energy, in contrast to a pixel in a 2D camera, which measures only accumulated energy.

Range to a reflective target in the lidar camera's field of view is inferred from the elapsed time between the emission of a pulse and the arrival of the pulse echo. For the camera to form a 3D image, range measurement must be carried out independently for each pixel. After the emission of a pulse, each pixel precisely records light intensity versus time for a finite duration. The net intensity is a sum of the pulse echo and undesirable contributions (i.e., noise) from various sources, including reflected sunlight and the electronics. Ideally, the pulse echo will stand out in the intensity data as a high, narrow peak whose time corresponds to the target range.

Echo arrival time measurement is a classic detection problem. When the echo signal is ab-

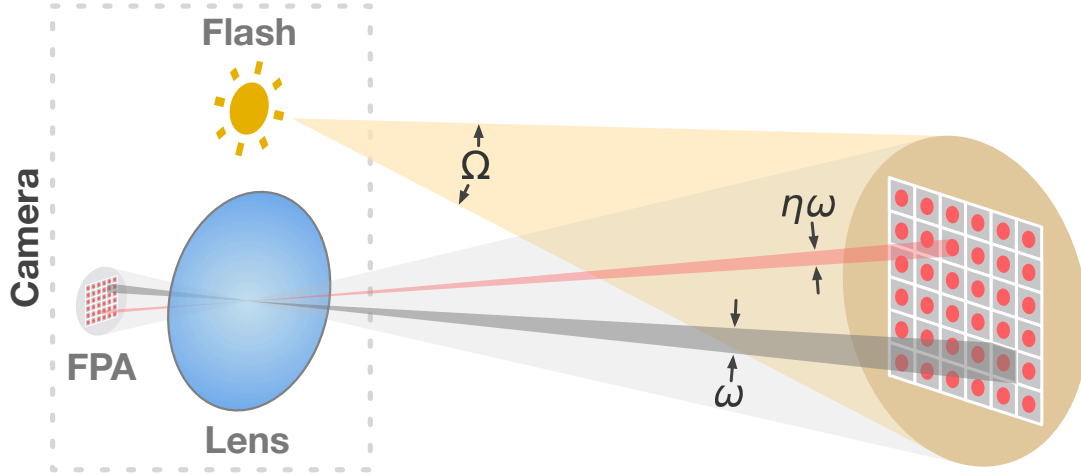


Figure 1: Architectural schematic of a flash lidar camera. The flash illuminator, lens, and FPA are colocated in a housing represented abstractly by the dashed box. It is assumed that the illuminator and the FPA subtend the same angular field of view with solid angle Ω far from the camera. The fields of view of the pixel and the pixel sensitive area (red) have respective solid angles ω and $\eta\omega$, where $\eta < 1$ is the *fill factor* of the sensitive area.

sent, the intensity data records random fluctuations of noise sources about some mean value. These fluctuations are drawn from a known probability distribution function. A significant local maximum in the intensity is one with a large enough amplitude that it is unlikely to have arisen as a random noise fluctuation. Detection involves finding an intensity peak above a certain threshold, where the threshold is chosen to maximize the probability of discovering the intended signal when it is present, under the constraint that the probability of a noise-only detection (i.e., false alarm) is below a certain tolerance.

A metric that positively correlates with the probability of detecting the signal of interest is the ratio of the signal strength to the magnitude of random fluctuations that could be misinterpreted as the signal. The next section presents this ratio in the context of flash lidar. The false alarm probability is not addressed in this report.

2 Signal-to-Noise Ratio

As mentioned above, 3D image construction requires that each pixel records intensity data following a pulse. Assume that the pulse power is constant for duration τ and that intensity is recorded at the pixel for a time t_g (the *gate time*) after the pulse. Furthermore, suppose that intensity is discretized as binned photon counts, the bin width equals the pulse duration, and echo photons are contained in a single bin. Figure 2 illustrates the series of measurements. It is appropriate to think in terms of photon counts because, as will be seen in Section 6, the echo intensity can be so low that each pixel must be capable of detecting and timing individual photons.

Let s and n be, respectively, the expected total number of signal photons and the expected number of noise photons per pixel per time bin. If the arrival-time bins are statistically independent, then the actual bin counts are drawn from a Poisson distribution, which has the property that the standard deviation is the square root of the mean. The standard deviation characterizes the spread of the measured values. In the echo bin, the mean and spread are $s + n$ and $(s + n)^{1/2}$. If s is large compared to the spread, then there is a high probability of

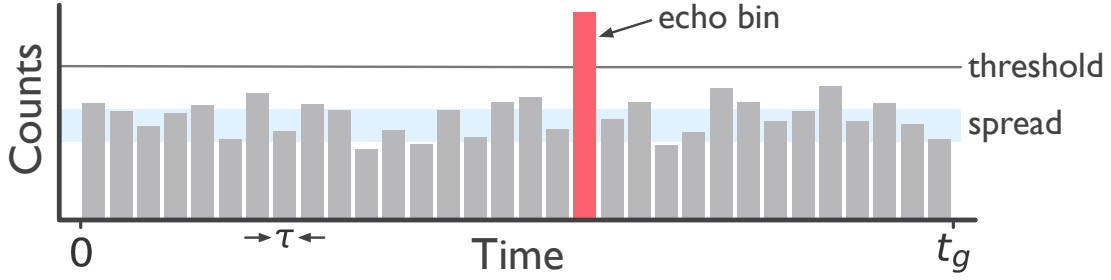


Figure 2: Binned intensity measurements. Light intensity is discretized as photon counts per bin of width τ over the observation interval of duration t_g following a single pulse. The red bin highlights the arrival time of the pulse echo.

observing a strong peak in the echo bin. The *signal-to-noise ratio*^[1] (SNR) measures the signal strength relative to the spread. For a single pulse, the SNR is

$$\text{SNR}_1 = \frac{s}{(s+n)^{1/2}} . \quad (1)$$

Under a wide range of practical conditions, the echo from a single flash lidar pulse is too weak to allow range measurement (see Section [6]). In fact, both s and n may be much less than 1. In this case, the signal can be amplified by accumulating bin counts over $M \gg 1$ pulses,^[2] which leads to a net SNR of

$$\text{SNR}_M = M^{1/2} \text{SNR}_1 . \quad (2)$$

A typical SNR threshold for robust detection is $\text{SNR}_{\text{th}} = 3\text{--}10$. If $\text{SNR}_1 \sim 1$, of order 10 to 100 to pulses are needed to achieve the detection threshold.

In the next two sections, expressions are derived for s and n as functions of flash lidar design parameters and external conditions. These results are then combined to yield the equation for the SNR that is used for tradeoff analysis.

3 Signal

The detected signal photon count per pixel after a pulse depends on the light energy received by the lens after the pulse reflects from a distant target. The received echo energy depends, in turn, on the total energy and angular spread of the emitted pulse, the target distance and reflectivity, the lens area, and the size and photon detection efficiency of each pixel. All of these elements combine to yield an expression for s . This section provides an expanded discussion of the system architecture as it pertains to the computation of the signal count.

Each pulse has total energy E spread over the illumination field of solid angle Ω . Here it is assumed that, far from the camera, Ω is approximately equal to the FPA field of view. If there were a large difference between the FPA and illumination solid angles, there would be significant waste in either the light output or FPA area. The total field of view is tiled by N pixels, each with a projected solid angle of $\omega \approx \Omega/N$. If only a fraction η of each pixel area is

¹The term *noise* can refer to either random measurement variations that may masquerade as the signal of interest, as in the definition of SNR, or to undesirable detector inputs. Hereafter, *noise* will refer to non-echo photon counts rather than statistical fluctuations.

²In the context of low-intensity time-of-flight measurement, multi-pulse aggregation is referred as time-correlated single-photon counting (TCSPC), a technique widely used in molecular fluorescence studies (e.g., Massa et al. [1997]; O'Connor [2012])

sensitive to light, the corresponding viewable solid angle is $\eta\omega$. In Fig. 1, gray squares and red circles in the projected FPA represent the pixels and light-sensitive regions, respectively.

The projected angular diameter of a pixel measures how finely the field of view is sampled by the FPA. If the pixel diameter is d and the lens has focal length f , geometric optics dictates that the pixel angular diameter is $\phi \approx d/f$, and thus the pixel field of view is $\omega \approx \phi^2 \approx (d/f)^2$. As long as the diameters of the pixel and pixel sensitive area are significantly larger than the wavelength of collected light, and the optics are ideal, it can be assumed that the lens focuses to a point on the FPA. Under these circumstances, ϕ can be treated as the *angular resolution* of the camera, which might be an independent design constraint.

To quantify the amount of reflected light received by a pixel, consider a planar target at range R from the camera oriented at normal incidence to the light from the illuminator. It is assumed here that the target is *resolved* in the sense that the target angular area is large enough to cover many pixels on the FPA. For a single pixel, the sensitive region is projected onto the target as a spot with area $\eta\omega R^2$. Light reflected from outside this spot will not be detected by the pixel in question. Of the light reflected from the spot, only rays within the cone subtended by the lens are detectable. If the lens has area A_l , the acceptance cone has solid angle A_l/R^2 .

The energy incident on the projected sensitive area of a single pixel is $\eta\omega E/\Omega$. If the surface is a diffuse, or Lambertian, reflector, then the fraction of the incident energy reflected into a unit solid angle about the surface normal is ρ/π , where ρ is the albedo.³ Let κ be the net fraction of energy incident on the lens that is detected by the pixel.⁴ The reflected pulse energy detected by a single pixel can now be written as

$$E_d = \frac{\rho\kappa\eta\omega A_l E}{\pi\Omega R^2} . \quad (3)$$

If the emitted pulse light is nearly monochromatic with wavelength λ , the energy per photon is $\epsilon = hc/\lambda$, where h is Planck's constant and c is the speed of light. It follows that the expected number of detected echo photons is

$$s = E_d/\epsilon . \quad (4)$$

Through eqs. (3) and (4), the simple expression for s spans much of the design space by combining the parameters of the illuminator, lens, pixel, FPA, and target.

Here and in what follows, the parameters ρ , κ , and η , appear only in the product $\rho\kappa\eta$, which is hereafter be denoted by α . The factor α is essentially the net photon detection efficiency. It encapsulates the total pixel detection efficiency $\kappa\eta$ and the target reflectivity ρ . Conversely, $1 - \rho\kappa\eta$ is the fraction of emitted photons that are lost when the illuminator and FPA fields of view are equal. In general, α will also incorporate losses due to atmospheric scattering and absorption, which depend on both range and wavelength; these effects are not considered here.

4 Background

Several classes of noise contribute to the net count n . For illustrative purposes, and because it is often dominant, the only noise considered here is reflected background light, which is any light reflected from the target that is not from the lidar illuminator. Sunlight is typically the

³<https://www.cs.cmu.edu/afs/cs/academic/class/15462-f09/www/lec/lec8.pdf>.

⁴This factor incorporates optical losses through the lens and the detector quantum efficiency, which depends on the detector type and the wavelength of light.

strongest background source in outdoor lidar applications, but vehicle headlights, street lights, or lighted signs may also be significant under certain circumstances.

Denote by B the background flux (power per unit area) on the target surface. This flux already accounts for a filter—placed adjacent to the lens or FPA—that permits only a narrow spectral band around the illuminator wavelength. For a single pixel, the projected photo-sensitive area receives energy $\eta\omega R^2\tau B$ over the pulse duration. Given this incident energy, the remainder of the analysis mirrors the derivation of the signal count. When background light is the only source of noise, the expected noise count per time bin is

$$n = \frac{\alpha\omega A_l \tau B}{\pi\epsilon} . \quad (5)$$

Notice that this result is independent of R , which implies that s/n decreases in proportion to R^{-2} as the target is moved away from the camera. Also, the background noise count depends explicitly on τ , while s does not.

5 Design Equation

Substitution of eqs. (3), (4), and (5) into eq. (2) yields, after some rearrangement,

$$\text{SNR}_M = M^{1/2} \left(\frac{\alpha\omega A_l}{\pi\epsilon} \right)^{1/2} \frac{E/\Omega R^2}{(\tau B + E/\Omega R^2)^{1/2}} . \quad (6)$$

This equation shows explicitly how flash lidar sensitivity depends on the system design and external factors. Through eq. (6), it is possible to identify key design tradeoffs or optimal configurations under external constraints.

Since the SNR is dimensionless, it may be advantageous to cast the right side of eq. (6) in terms of dimensionless variables. To do this, note that there is a range R_e for which $s = n$:

$$R_e = \left(\frac{E}{\Omega\tau B} \right)^{1/2} . \quad (7)$$

At short ranges compared to R_e , $\text{SNR}_M \propto R^{-1}$, while at long ranges, $\text{SNR}_M \propto R^{-2}$. In terms of $\xi = R/R_e = (s/n)^{-1/2}$, eq. (6) becomes

$$\text{SNR}_M = M^{1/2} n^{1/2} \xi^{-1} (1 + \xi^2)^{-1/2} . \quad (8)$$

Equation (8) better highlights the dependence of the SNR on range, and it isolates key design parameters in R_e and n .

From a detection perspective, eq. (6) suggests that it is always better to have more efficient detector elements, a lens with a smaller focal length and larger area for a given pixel diameter,⁵ a higher target reflectivity, and a shorter pulse. The net SNR always increases with E , although the rate of increase depends on R when all other parameters are fixed.

The dependence of SNR_M on the pulse duration τ is a bit more subtle. At fixed energy, SNR_M increases with decreasing τ because fewer background photons are counted in the echo bin. However, if pulse power E/τ is held constant, then SNR_M decreases as the pulse duration is shortened.

One common use case for eq. (6) is to assess how certain parameters scale when others are varied at a fixed SNR. That is, assume that the net SNR equals some detection threshold,

⁵This is equivalent to saying the lens should have a large *f*-number.

Variable	Description	Value	Unit
E	Total light energy per pulse	50	μJ
τ	Pulse duration	5	ns
α	Photon detection fraction	0.01	dimensionless
Ω	Total field of view	3000	sq. deg
ω	Pixel field of view	0.01	sq. deg
A_l	Lens area	7	cm^2
ϵ	Photon energy	2×10^{-19}	J

Table 1: Fiducial parameter values for long-range flash lidar.

and discern the relationship between the remaining variables. As a specific and relevant example, consider the situation where $s/n \ll 1$ ($R \gg R_e$) and parameters pertaining to the optics, FPA, background flux, target reflectivity, and wavelength are constant. In other words, the hardware package and target surface are unchanged, while the target range and the pulse energy, duration, and count are allowed to vary. The resulting scaling relation is

$$\left(\frac{E}{E_0}\right)\left(\frac{M}{M_0}\right)^{1/2}\left(\frac{\tau}{\tau_0}\right)^{-1/2}\left(\frac{R}{R_0}\right)^{-2} = 1, \quad (9)$$

where the subscript 0 denotes a reference value. For example, when M and τ are fixed, the pulse energy required for detection increases by a factor of 100 when the range is 10 times larger than the reference value. Alternatively, when E and τ are constant, a 10000-fold increase in the pulse count is needed to achieve detection at 10 times the range. Whether it is more advantageous to increase the energy or the pulse count depends on the allowed operating envelope.

6 Numerics

Table 1 shows a set of fiducial parameter values for a flash lidar system capable of detecting targets at $\gtrsim 100$ m. It is assumed in this table and in what follows that $\lambda = 1 \mu\text{m}$.

The key parameter that is not expressed in Table 1 is the solar background flux B . An approximate *upper limit* to the solar background intensity at the Earth's surface is given by

$$b(\lambda) = 8000 \lambda^{-5} (e^{2.5/\lambda} - 1)^{-1} \text{ W m}^{-2} \mu\text{m}^{-1}, \quad (10)$$

where λ is measured in microns. This spectrum applies for direct sunlight at midday, middle latitudes, and at wavelengths not near any major absorption bands. When sunlight passes through a narrow bandpass filter of width $\Delta\lambda \ll \lambda$, the detectable flux is $B(\lambda) \approx b(\lambda)\Delta\lambda$. When $\lambda = 1 \mu\text{m}$, the maximum filtered flux is

$$B(1 \mu\text{m}) \approx 7 \left(\frac{\Delta\lambda}{0.01 \mu\text{m}} \right) \text{ W m}^{-2}. \quad (11)$$

Values appropriate for indirect sunlight and average times of day may be significantly less than this upper bound. Outside of absorption bands, filtered solar background fluxes less than $\sim 1 \text{ W m}^{-2}$ are more likely to be encountered in clear daytime conditions. A background flux of 1 W m^{-2} will be adopted as the fiducial value.

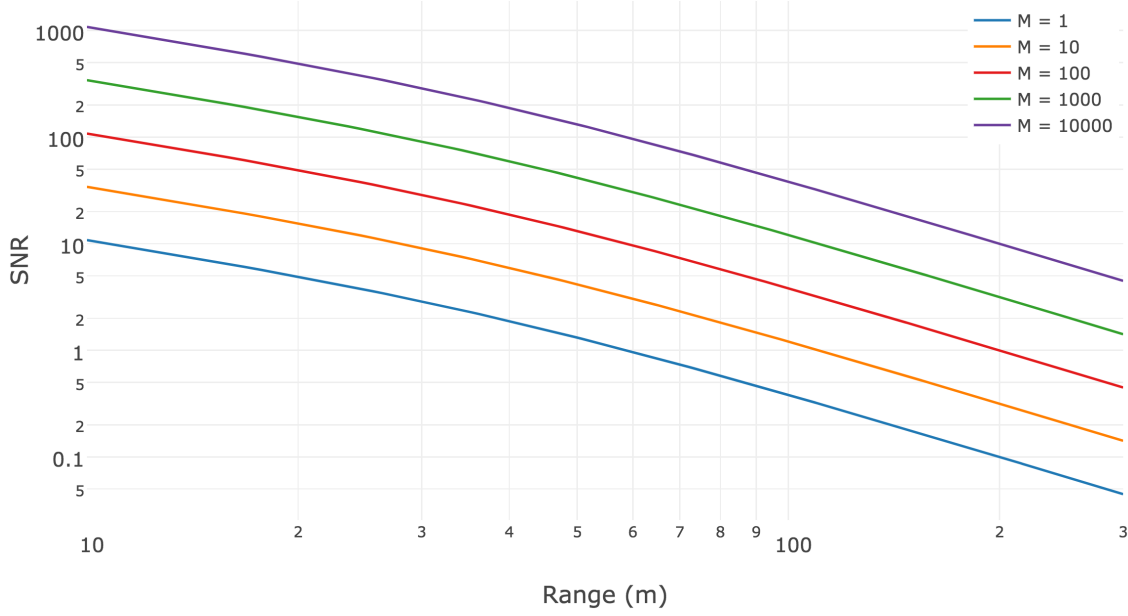


Figure 3: SNR for fiducial parameter values.

Substitution of the above values in eqs. (4) and (5) yields

$$s \approx 1.26 \left(\frac{R}{100m} \right)^{-2} \quad (12)$$

and

$$n \approx 9.57 . \quad (13)$$

At ranges of $\gtrsim 100$ m, both the signal and the noise counts are expected to be very low under the reference model. The corresponding SNR is

$$\text{SNR}_M \approx M^{1/2} \frac{1.26 R_{100}^{-2}}{(9.57 + 1.26 R_{100}^{-2})^{1/2}} , \quad (14)$$

where $R_{100} = R/100$ m. Figure 3 shows eq. (14) for five different values of M . For the reference parameter values, detection is possible when $M \gtrsim 100$. Detection is unlikely for ranges beyond 300 m.

References

- Massa, J., Wallace, A., Buller, G., Fancey, S., and Walker, A. (1997). Laser depth measurement based on time-correlated single-photon counting. *Optics letters*, 22(8):543–545.
- O'Connor, D. (2012). *Time-correlated single photon counting*. Academic Press.